Statistical Physics

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1 Basics of statistics

Imagine we want to describe a system of a huge number of particles which is a *closed system*, meaning it does not exchange particles or interacts in any way with the world around it. The fundamental problem we face when dealing with that many particles is not that we could not solve the equations of motion for all these particles in principle but that it is just not useful to attempt this. The sheer magnitude of equations you would need to solve would take any given period of time.
However there are certain quantities that arise from the percieved chaos of the many particles which we can extract using statistics. These quantities *emerge* from a huge number of particles and do not carry meaning for a small number of particles.

1.1 Phasespace

Lets imagine a mechanical system with s degrees of freedom. For a three dimensional system this would be

$$s = 3N\,, \tag{1.1}$$

where N is the number of particles. That means we have q_i variables where $i = 1, ..., s$. To describe the state of the system completely we need to also know the velocity of every particle, $\dot{q}_i$. Normally we use the momentum $p_i = m\,\dot{q}_i$ instead of $\dot{q}_i$. We can introduce so called phasespace to describe the system. Phasespace is an imagined space that totally describes the state of the system. The axis of phasespace consist of the location variables and the momentum variables. If we imagine a mass on a spring for example which can only move along one axis, we can even visualize its phasespace (figure 1.1). Be aware that there are multiple ways of defining the phasespace. In our definition, we describe the whole system as one combined point in phase space (This is called a Gamma-Space). In other formulations you could describe every single particle in phasespace individually but this is not what we are doing here. As the system develops in time it forms a trajectory in phase space which we call *phase trajectory*.

1.2 Subsystems

We can define any system within our large system as a subsystem of particles. These so called *subsystems* do also carry a large ammount of particles but are small compared to the larger system. Nevertheless there is a huge difference to the larger system. The subsystem is not a closed system. Particles can freely enter and leave the subsystem wheareas the larger system is a closed system and does not interact with outside systems.

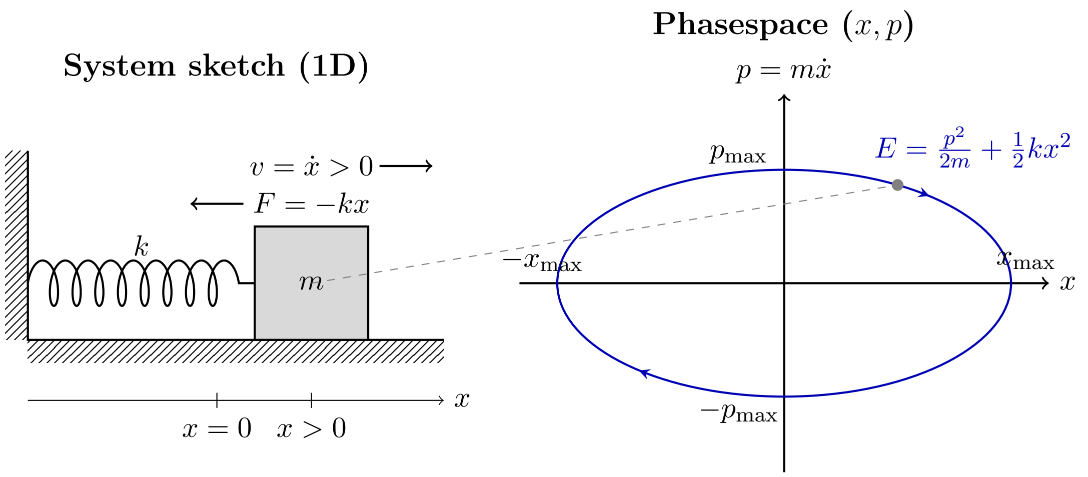

Figure 1.1: Visualization of the phase space for a one-dimensional harmonic oscillator (mass on a spring) that moves frictionless over a surface. The system is located at $x > 0$ and moving to the right ($v > 0$), which corresponds to the marked point in the phase space.

Key Concepts & Formulas

Closed system: Does not exchange particles or interacts in any way with the world around it.
Phasespace (Γ-Space): A $6N$-dimensional imagined space that completely describes the mechanical state of a system with N particles.
Phase trajectory: The trajectory that the system takes, as it moves through phasespace

Bibliography

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